

Ben-Gurion University of the Negev

Faculty of Natural Sciences | Department of Computer Science

Research Proposal for M.Sc. Thesis

Safe Constrained Multi-Agent Reinforcement Learning with World Models

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Abstract

Constrained multi-agent reinforcement learning is usually solved reactively: a policy learns to avoid a failure only after the environment has produced one. Learned world models offer another route, because most policy updates can be taken against imagined rollouts instead of collected ones. We propose *Cost-Aware MAMBA* (CA-MAMBA), which augments a multi-agent recurrent state-space model with per-agent cost prediction, trains factorized actors on imagined reward and cost advantages, and calibrates one team-level Lagrange multiplier from measured environment cost. We characterize when this transfers: bounded one-step cost and transition error imply an explicit finite-horizon bound on the gap between imagined and real cost, which fixes the conservative margin an imagined constraint must carry. Empirically, at a matched interaction budget CA-MAMBA lowers casualty cost on all 24 shared StarCraft configurations and wins more often on all 24, while the model-free baseline wins no episode on 22 of them; the ordering largely persists when that baseline is given fifty times more data. On the SafePO velocity benchmark, at a matched budget it holds mean cost under the limit while earning 2.8 to 12 times the reward of both MACPO and MAPPO-Lagrangian in all three environments.

1 Scientific Background

1.1 Constrained Multi-Agent Reinforcement Learning

In cooperative multi-agent reinforcement learning (MARL), a team of agents must learn decentralized policies that maximize a shared reward signal under partial observability. Adding safety constraints turns this into a *constrained* Dec-POMDP [1]: the team must keep an expected cumulative cost below a given budget d while maximizing return.

Constrained policy optimization [2] controls expected cost through trust-region updates. MACPO and MAPPO-Lagrangian carry this line into multi-agent robot control [7], both built on MAPPO [8]. Related formulations include CMIX, which adds peak to average constraints in a value-factorized learner [9], and risk-sensitive methods such as RMIX [10]. Every update in this family is computed from environment trajectories—exactly the cost the method is trying to avoid paying.

1.2 Model-Based Reinforcement Learning and World Models

Dreamer [3] improves a policy through rollouts of learned recurrent latent dynamics, a line since scaled across domains [4]. SafeDreamer adds cost prediction, separate reward and cost critics, and Lagrangian optimization to this single-agent framework [6]. MAMBA adapts latent world models to cooperative partial observability by aggregating other agents’ latents and actions through attention [5].

None of these three systems, by itself, addresses the multi-agent constrained setting: Dreamer and SafeDreamer are single-agent, and MAMBA has no cost channel. The composition of these components—and the questions it raises about cost aggregation, communication, and model error in the constraint—is the gap this research addresses.

2 Research Goals

The research aims to answer the following questions:

1. **Cost-aware world model for MARL.** Can a multi-agent world model be extended with per-agent cost prediction so that a factorized policy trained on imagined rollouts achieves lower realized safety cost than model-free constrained baselines at a matched interaction budget?
2. **Theoretical transfer guarantees.** Under what conditions does a constraint enforced in imagination transfer to the real environment? Specifically, can bounded one-step cost-head error and transition error yield an explicit finite-horizon bound on the gap between imagined and realized cost?
3. **Communication requirements.** How much inter-agent communication is necessary for the constrained policy to function? Is this a property of the domain rather than of the framework, and can it be measured by varying the attention neighborhood at execution time?
4. **Generalization across cost structures.** Does the approach generalize across different cost signals (sparse/discrete casualty cost on SMAC vs. continuous velocity constraints on MAMuJoCo) and action spaces (discrete vs. continuous)?

3 Research Methodology

3.1 Constrained Dec-POMDP Formulation

We formulate the problem as a finite-horizon constrained Dec-POMDP $\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_i\}, P, R, \{C_i\}, \{\mathcal{O}_i\} \rangle$ where each agent i has a local cost C_i and the team cost is the mean $c_t = \frac{1}{N} \sum_i c_t^i$. The joint policy factorizes as $\pi_\phi(\mathbf{a}_t \mid \boldsymbol{\tau}_t, \mathbf{m}_t) = \prod_{i=1}^N \pi_{\phi_i}^i(a_t^i \mid \tau_t^i, m_t^i)$, where m_t^i is a message from attended teammates. The objective is to maximize discounted return J_R subject to $J_C(\pi_\phi) \leq d$.

3.2 Cost-Aware Multi-Agent World Model (CA-MAMBA)

The world model is a recurrent state-space model (RSSM) with per-agent deterministic state h_t^i and stochastic state z_t^i . Three heads decode each latent state: observation (for representation learning), reward, and *cost*—the latter is our addition. Each cost head regresses agent i ’s own cost c_t^i ; the imagined team cost is the mean $\hat{c}_t = \frac{1}{N} \sum_i \hat{c}_t^i$.

Inter-agent communication is realized via a multi-head Transformer encoder that aggregates teammates’ previous actions and stochastic latents, parameterized by an attention neighborhood $\mathcal{N}_i$. Setting $\mathcal{N}_i = \mathcal{I}$ gives all-to-all communication; $\mathcal{N}_i = \{i\}$ yields communication-free execution.

3.3 Imagined Actor–Critic and Lagrangian Dual

A Lagrange multiplier $\beta \geq 0$ relaxes the constraint. Two separate critics estimate imagined reward and cost returns, yielding a combined advantage $A_u^\beta = A_u^R - \beta A_u^C$. The actor maximizes a clipped surrogate objective over imagined samples. The multiplier is updated from *measured* environment cost: $\beta_{k+1} = [\beta_k + \alpha_\beta(\bar{C}_k - d)]_+$, creating a hybrid where the actor uses imagined cost but the dual uses real cost.

3.4 Theoretical Analysis

Theorem 1 (Conditional Cost Transfer). *Under bounded one-step cost error ϵ_c and transition error ϵ_P (in total variation), the gap between imagined and real discounted cost over horizon H_{im} satisfies:*

$$|J_{C,H_{\text{im}}}(\pi) - \hat{J}_{C,H_{\text{im}}}(\pi)| \leq \epsilon_c \sum_{t=0}^{H_{\text{im}}-1} \gamma^t + c_{\max} \epsilon_P \sum_{t=0}^{H_{\text{im}}-1} t \gamma^t.$$

This bound identifies cost-head error as entering linearly in the horizon and transition error quadratically, naming which error source to reduce first. We further show that the measured-cost dual update has the correct sign (it is projected stochastic descent in β) but cannot repair an actor gradient taken through a cost-blind model.

3.5 Experimental Plan

Environments:

- *StarCraft Multi-Agent Challenge (SMAC)*: 12 maps with casualty cost (sparse, discrete, irreversible) and collision cost (dense, continuous, reversible).
- *MAMuJoCo (SafePO)*: Three velocity-constrained locomotion tasks with continuous actions (Ant 2×4, Ant 4×2, HalfCheetah 2×3).

Baselines: MACPO-Discrete (SMAC), MACPO and MAPPO-Lagrangian (MAMuJoCo), all from the SafePO repository.

Ablations:

- Communication: full ($\mathcal{N}_i = \mathcal{I}$) vs. self-only ($\mathcal{N}_i = \{i\}$) at execution time.
- Multiplier learning rate α_β : effect on cost–reward trade-off.
- Cost machinery removal: unconstrained world-model learner to isolate the cost mechanism’s contribution.

Metrics: Win rate and shaped score (SMAC); episodic reward and cost (MAMuJoCo); cost-head one-step prediction error; multiplier dynamics.

3.6 Preliminary Results

Preliminary experiments confirm the approach. At a matched budget of 100,000 interactions, CA-MAMBA has lower mean casualty cost than MACPO-Discrete in 24/24 SMAC configurations and higher win rate in 24/24; the baseline wins zero episodes on 22 of 24 configurations. On MAMuJoCo at a matched 1M budget, CA-MAMBA earns $2.8\text{--}12\times$ the reward of both baselines in all three environments while holding cost under the limit. The cost head’s measured one-step error is at most 0.004 ($\sim 3\%$ of peak cost), making the transfer bound quantitative.

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